

Interface Capability Brief

Medical Device HL7 → EHR Demo · Stakeholder overview · 2026-08-05

1. Executive summary

Demo slice: two mocked bedside devices send HL7 ORU observations into PilotFish over LLP; EIP validates and forwards HL7 2.x to a mocked EHR LLP endpoint.

2. Who this is for

- Sources (mocked): 1. Bedside Vital Signs Monitor (VITALMON) — HR, SpO2, NIBP 2. Continuous Glucose Monitor (CGM01) — interstitial glucose - Destination: Hospital EHR mock (EHR MOCK) via HL7 LLP - Demo vs production: Demo only

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: HL7 LLP (HL7TCPLListener) host 2580
- Format: HL7 v2.5.1 ORU^R01
- Identity: MSH-10; device id in MSH-3
- Samples: samples/VITALS_ORU_hl7, samples/CGM_ORU_hl7

Integration highlights

FQCNs from PilotFish_Documentation tracker + PilotFish_V2 format-hl7 / modules.conf. Runtime V1 on pilotfish-eip:23R1.

Stage	Module	Notes
Listener	HL7TCPLListener	:2580, PilotFish, async auto-ACK
Archive	FileWriteProcessor	inbound
Validate	HL7ValidationProcessor	ThrowException=true
Audit	FileWriteProcessor	ehr-outbound
Router	NullRoutingModule	All Targets (raw HL7)
Transport	HL7SimpleTransport	ehr-mock:2581

4. Honest boundaries (not claimed yet)

- Production hardening (HA, disaster recovery, full audit workflows) is outside this demo brief
- Anything not listed in §3 should be treated as future work unless confirmed in DESIGN.md

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

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- Exercise route “1 - Process Device ORU LLP” from the Web UI or documented smoke path
 - Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

6–8. State / validation / side effects

- Fail-closed HAPI parse before EHR send - Async AA (same 23R1 sync-ACK limitation as lab demo) - No SQL; evidence is files + ACK - Scope excludes PDF reports, FHIR, demographics enrollment from the case-study diagram

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- Fail-closed HAPI parse before EHR send - Async AA (same 23R1 sync-ACK limitation as lab demo) - No SQL; evidence is files + ACK - Scope excludes PDF reports, FHIR, demographics enrollment from the case-study diagram

3. Inbound contract

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Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Process Device ORU LLP: runs steps such as Device HL7 LLP Inbound, Archive Device ORU, HL7 Parse Validation, Audit EHR Outbound.

7. Security & trust posture

- Inbound clinical/FHIR payloads can be validated before persistence
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — Case study broader than demo — Yes — ORU LLP slice + 2 devices
- Risk noted in design: Med — Async ACK before EHR durable — Yes
- Risk noted in design: Low — Device sims are software emitters, not real IoT — Yes

8. Seeing it work

- - EIP 8100, Web UI 8101, device LLP 2580, EHR mock 2581
- docker compose up -d --build
- Open <http://localhost:8101/> and use Simulate reading for each device.

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- DeviceHL7EHR_V2_Route_Diagrams.pdf — detailed PilotFish route wiring diagrams
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.